Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. We trained each model for 100 epochs with a batch size of 128.

2 Results

Figure 1 shows the f1 score during training.

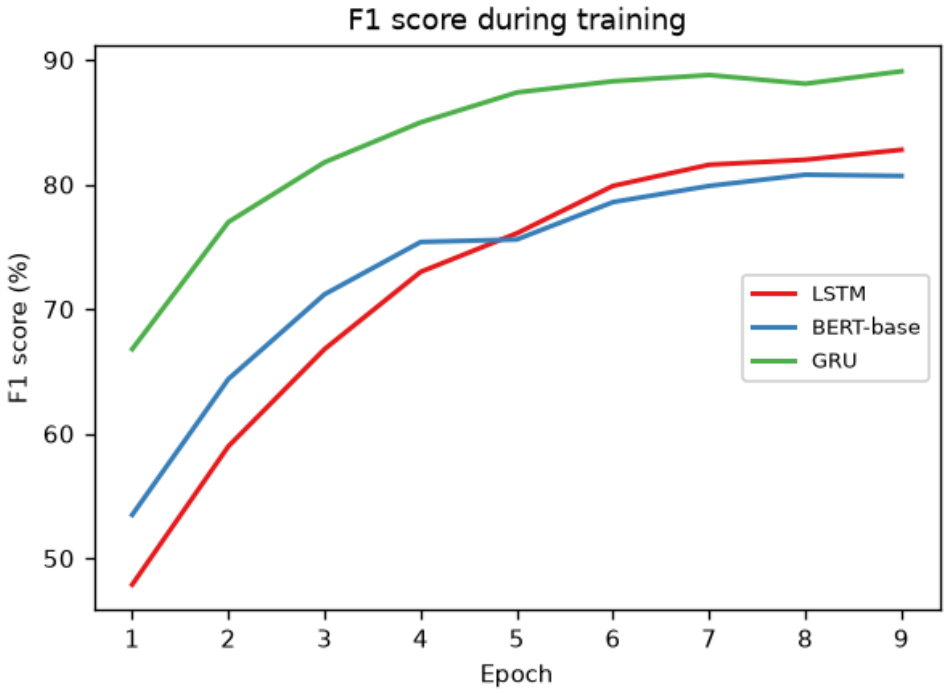


Figure 1: F1 score during training for the models in this experiment.

As shown in Figure 1, from epoch 4 to epoch 7, the F1 score of LSTM improves by a relative 13.0%. At epoch 7, LSTM is 2 points better than BERT-base (Figure 1). As shown in Figure 1, GRU reaches 88.2% F1 score at epoch 9. As shown in Figure 1, BERT-base attains a F1 score of 89.0% at epoch 7.

Figure 2 shows the validation accuracy during training.

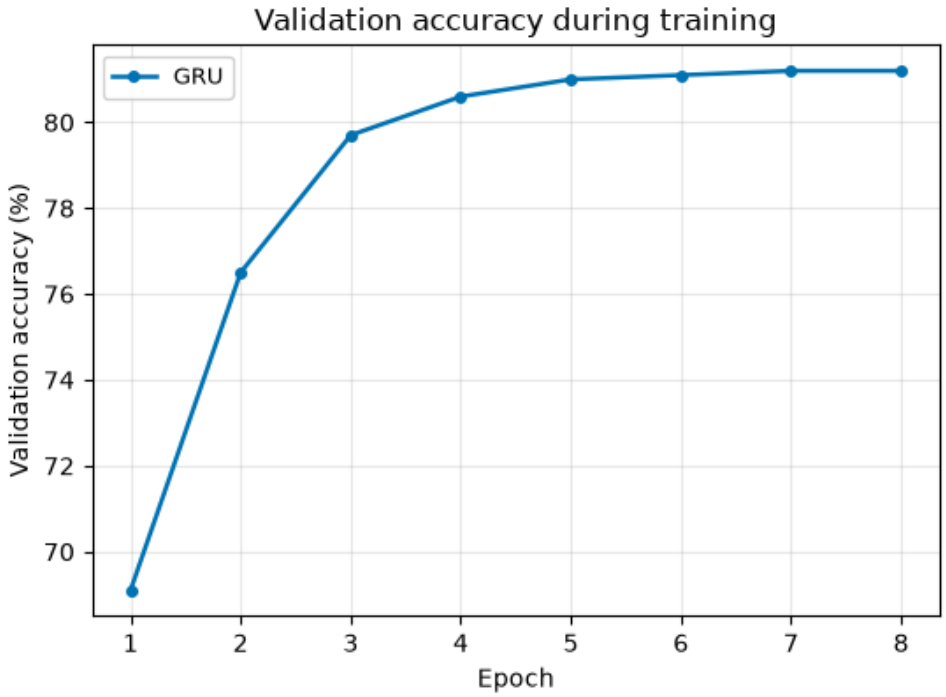


Figure 2: Validation accuracy during training for the models in this experiment.

GRU reaches 81.2% validation accuracy at epoch 8. At epoch 4, GRU achieves about 80% validation accuracy. As shown in Figure 2, GRU scores about 4 points higher on 4 than on 2.

Figure 3 shows the training loss.

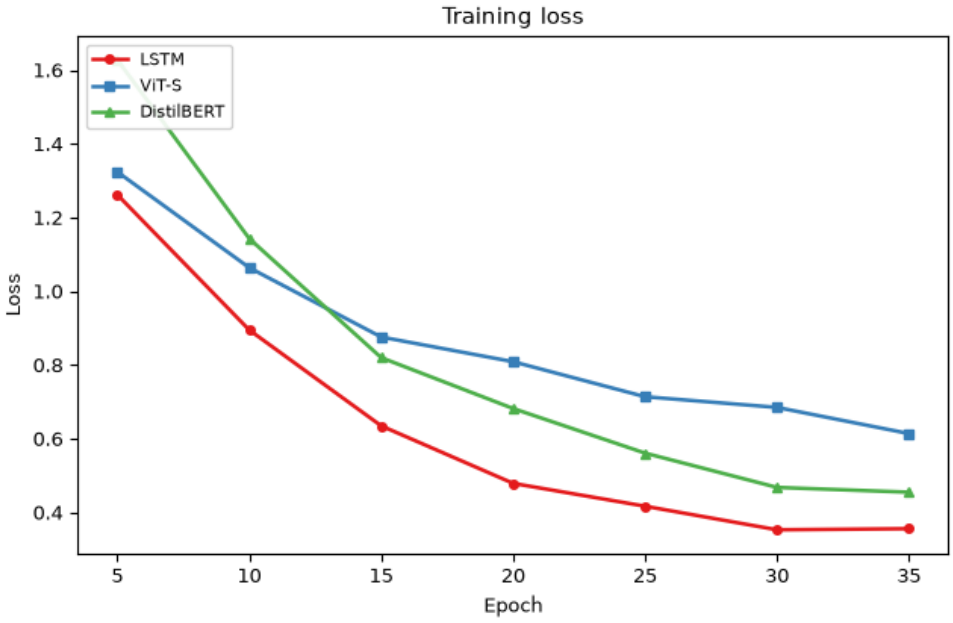


Figure 3: Training loss for the models in this experiment.

DistilBERT has a loss of 0.483 at epoch 35. LSTM has a loss of -0.222 at epoch 20 (Figure 3). DistilBERT has a loss of 0.820 at epoch 15.

Figure 4 shows the f1 score during training.

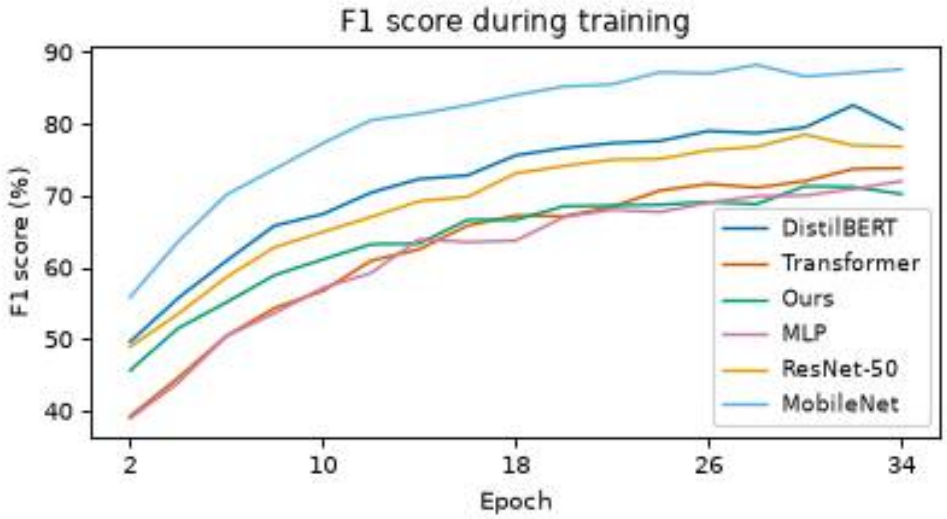


Figure 4: F1 score during training for the models in this experiment.

At epoch 10, MobileNet is 47.7 points better than Transformer. MobileNet outperforms ResNet-50 by 12.8 percentage points at epoch 16. From epoch 8 to epoch 22, the F1 score of DistilBERT improves by a relative 1.0%.

3 Discussion

On IMDB, XGBoost obtains an accuracy of 82.0%. We note that the training was stable for all models.